

SMART INDIA HACKATHON 2026



TITLE PAGE

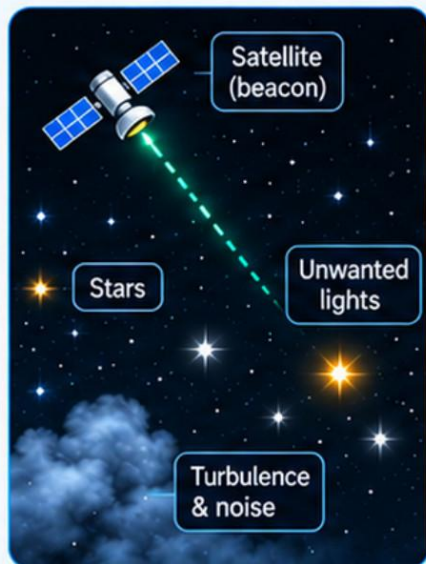
- Problem Statement ID – 26169
- Problem Statement Title – AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile FSOC Terminals
- Theme – Smart Automation
- PS Category – Software
- Team ID – —
- Team Name – ZeroDrift



The eyes and neck of a laser communication terminal

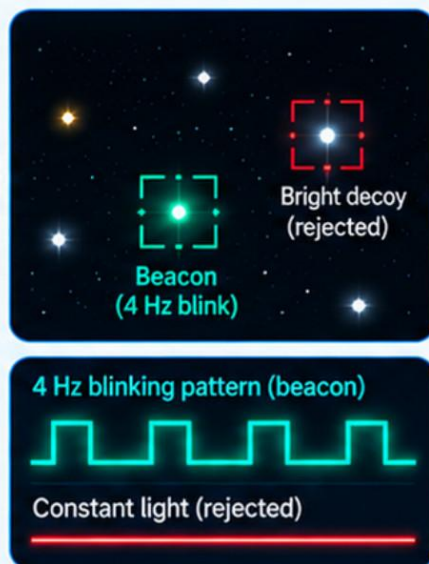
1 1. Simulate

Virtual sky with beacon, stars, noise, turbulence and unwanted lights.



2 2. Verify

Detect the beacon and reject bright decoys using the 4 Hz blinking signature.



3 3. Track & Predict

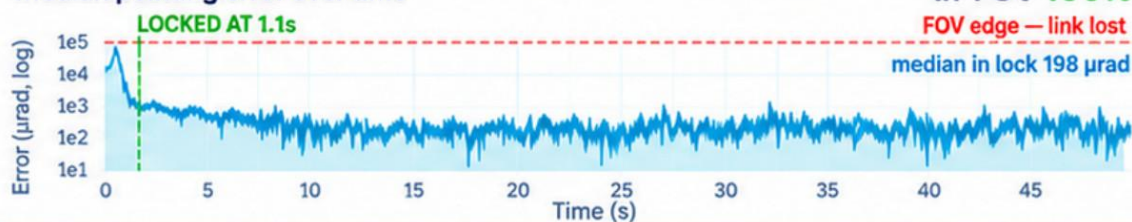
Predict motion and continuously steer the virtual camera to keep the target centered.



Console locked on the beacon — the same view the live demo link opens



Median pointing error over time



CORE



Python 3.11 · NumPy · OpenCV

INTERFACE



PySide6 console · pyqtgraph live plots

ORBITS · AI



real ISS TLEs · Neural Network

QUALITY



pytest suite · GitHub Actions

SHIPPING



Netlify



CORE TRACKING PIPELINE

Simulate → Capture → Detect → Confirm → Output

ORBITAL
DATA & AIISS (International
Space Station)

- Using real ISS TLEs with SGP4 for position propagation.

NN (Neural
Network)

- Custom model to verify true ISS detections and reject decoys.

1 VIRTUAL WORLD

Simulate the space environment



- Generate a realistic sky with stars, clutter and decoys using real ISS orbital data (TLEs) and UAV/LEO trajectories.
- Render high-fidelity imagery at **9.2 MPx / s**.

2 VIRTUAL CAMERA

Capture realistic frames



- A pan-tilt gimbal camera simulates real hardware constraints (slew limits, sensor noise, etc.).
- Captures frames at **30 frames / s** with 40 ms latency per frame.

3 PERCEPTION

Detect and verify candidates



- Each frame is processed using a top-hat + matched filter to enhance bright objects.
- Compute centroids and use an AI verifier (Neural Network) to filter true targets.
- 0 – 6 candidates / frame.**

4 DECISION

Confirm the target



- A multi-target tracker associates detections over time.
- Use blink-signature analysis and an IMM Kalman filter to estimate motion and identity.
- When the track matches the ISS signature, we confirm **1 beacon**.

5 OPERATOR OUTPUTS

Visualize, report and export



- A live GUI shows tracking status, estimated position and telemetry.
- Operators can view auto-generated performance reports and export demo videos for judges and operators.

Sends continuous pan/tilt pointing commands (30 commands / s).

CONTROL MODULE

Point the camera and run the search strategy



Smith Predictor

Compensates for 40 ms camera latency.



Velocity Feed-Forward

Helps the camera follow the predicted path.



Spiral Search States

Ensure efficient coverage when the target is not in view.



Control Loop

Runs at ~ 30 commands / s, continuously updating camera pointing.



ACQUISITION STATE MACHINE

Manages the overall tracking behavior



SEARCH

Scan predicted region for potential targets.



ACQUIRE

Detect candidates and validate them using the perception pipeline.



TRACK

Maintain a stable lock on the confirmed target and continuously refine the estimate.



COAST

Keep tracking with minimal control when the target is stable.



REACQUIRE

Recover from loss (even with decoy or off-target deviation).

If the target is lost, transition back to SEARCH



FEASIBILITY - PROVEN



Fully working today: 73/73
automated tests pass



64-run randomised Monte-Carlo
campaign: 100% acquisition; zero wrong locks



Standalone Windows/Linux
app builds in one click



Tracker uses 8-21 ms
of each 33 ms frame on two CPU cores —
no GPU, no cloud, no dataset



CHALLENGES & RISKS



Heavy turbulence
Dims and shakes the beacon



Bright decoys / sun glints
Can mimic the beacon



Mount latency
Destabilises fast tracking loops



Sharp UAV manoeuvres
Break simple trackers



OUR STRATEGIES



Coast-through-fade
+ automatic re-acquisition spiral



Blink-signature hard gate
Star scores zero — fooled 0 times in 64 runs



Smith predictor
Cancels the measured 40 ms delay



Dual-model IMM filter
Two motion models re-weighted every frame
catch sharp manoeuvres



Worst case across all 64 randomised runs: slowest acquisition 4.0 s, lowest lock retention 91.6%;
even a 0.53x-dim beacon under 1.74x turbulence acquired in 2.8 s. Every number regenerates from one command — fully reproducible.



STRATEGIC

Advances India's next-generation space & defence communications — FSOC pointing algorithms developed sovereignly, without importing optical test benches.



ECONOMIC

A laptop replaces lakhs of rupees of cameras, gimbals and optics — the entire develop-test-validate cycle for FSOC pointing runs in pure software, at zero recurring cost.



RESEARCH & EDUCATION

Any student or ISRO lab can develop, test and compare PAT algorithms instantly — reproducible scenarios, honest ground-truth scoring, open source.



SCALABLE & GREEN

Same engine covers satellite, UAV, ground and maritime links. Pure software — no consumables, no lab energy, no e-waste in the development cycle.



Target audience: ISRO / IN-SPACe optical-communication teams, DRDO, IITs & engineering colleges, and India's growing space-tech startups.



The number that matters: on a tracked ISS pass, coarse stage + modelled fine stage closes the optical link not asserted 99.3% of the time vs 14.7% without measured,





Research foundations

- 1 **Kaymak et al.**, "A Survey on Acquisition, Tracking and Pointing Mechanisms for Mobile FSO," IEEE Comm. Surveys & Tutorials, 2018. [\[link\]](#)
 - 2 **Kaushal & Kaddoum**, "Optical Communication in Space: Challenges and Mitigation Techniques," IEEE Comm. Surveys & Tutorials, 2017. [\[link\]](#)
 - 3 **Blackman & Popoli**, Design and Analysis of Modern Tracking Systems — IMM filtering, CFAR detection, track management. [\[link\]](#)
 - 4 **Bar-Shalom et al.**, Estimation with Applications to Tracking and Navigation — Kalman/IMM theory. [\[link\]](#)
 - 5 **Smith, O.J.M.**, "Closer Control of Loops with Dead Time" — the Smith predictor, 1957. [\[link\]](#)
 - 6 **Vallado & Crawford**, "SGP4 Orbit Determination" — AIAA 2008; real ISS TLE propagation. [\[link\]](#)
- Andrews & Phillips**, Laser Beam Propagation through Random Media — turbulence & scintillation models. [\[link\]](#)



Our work — open and verifiable



Repository: github.com/pranshukesavmishra/SIH-2026 [↗](#)



Live demo (replay console): zerodrift-fsoc-pat.netlify.app [↗](#)



Inside: full source (5,000+ lines, documented) · 73 automated tests · 6 validated scenarios · Monte-Carlo campaign logs · technical report · user manual · demo video.



Reproducibility: every figure in this deck regenerates from one command; every random draw is logged and replayable.



Why ZeroDrift wins on this problem



All six mandatory deliverables already exist today



Judged numbers, honestly measured against hidden ground truth



AI where it earns its place — benchmarked against classical methods



Compared with existing approaches

Feature	 ZeroDrift	 MATLAB sim	 Traditional
 Zero hardware cost No special equipment required	✓	✓	✗
 Real-time closed loop + live console See it run. Live.	✓	✗	✓
 Decoy & blink-signature discrimination Robust against spoofing	✓	✗	✗
 Open source · one-command reproducible Run with a single command	✓	✗	✗